

250+ CHEMISTRY & PHYSICS JAMB POINTS BY PHAGNOXEDUTECH

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CHEMISTRY

✚ Reddish brown colouration in soil is caused by what Iron ore ?

=> Haematite

✚ The most abundant Iron ore is?

=> Haematite

✚ The main component for the alloy permalloy is__?

=> Iron and Nickel i.e Fe & Ni

✚ Zinc(II) ion and scandium (III) ion are colourless due to complete empty electrons in their D-subshell respectively

✚ The enthalpy of neutralisation of HCl and NaOH is?

=> 57.3kJ/mol

✚ HF is a weak acid

✚ The basicity of CH_3COOH is 1

✚ Corrosion of iron is called rust

✚ The chemical formula for Rust is $\text{Fe}_2\text{O}_3 \cdot \text{XH}_2\text{O}$

✚ Hydrated iron (III) oxide is known as Rust

✚ In the electrolytic purification of impure metal, the metal is the anode

✚ The following are the four quantum numbers

Principal quantum numbers (n)

Azimuthal quantum numbers (L)

Magnetic quantum numbers (mL)

Spin quantum numbers (s)

✚ Azimuthal quantum numbers is also known as subsidiary quantum numbers

✚ Azimuthal quantum numbers consists of
S - sharp

P - principal

D - diffuse

F - fundamental

✚ Helium is preferred to hydrogen in filling balloons because it's inflammable (non-combustible)

✚ The longest period in the periodic table is 6 or 7 . If in the exams you see both , pick 6



✚ The production of alkanoate from the reaction of carboxylic acid and Alkanol is called....?

=> Esterification

✚ The reverse of Esterification is called....?

=> Hydrolysis

✚ The mild oxidation of primary alkanol yields/gives alkanal / aldehyde

✚ The complete oxidation of primary alkanal gives alkanoic acid or carboxylic acid

✚ The simplest aldehyde is methanal also known as formaldehyde (HCHO).

✚ The oxidation of secondary alkanol gives alkanone or ketones .

✚ The simplest alkanone is propanone also known as acetone

✚ What is the functional group in soap? It's the carboxylate group ($-\text{COO}-\text{Na}^+$)

✚ What is the IUPAC name of the by-product?
=>Propane-1,2,3-triol.

✚ Why is NaCl added in the saponification reaction?

=> To decrease the solubility of the soap so it can be collected , this is known as Salting out.

✚ What's the functional group in soapless detergent?

$\text{SO}_3^- \text{Na}^+$ also known as the hydrophilic head

✚ The water loving part of detergent is ...?
 $\text{SO}_3^- \text{Na}^+$

✚ Soapless detergent aren't biodegradable except it contains straight chain alkyl group.

✚ What's chemical name for scum ?
=> Calcium stearate

✚ The chief component of Gun powder is KNO_3 also known as Saltpetre

✚ NaNO_3 is also known as chile saltpetre

✚ Sucrose is a non reducing sugar

✚ Table sugar is sucrose

✚ The sweetest sugar is fructose

✚ The chemical formula for water glass is Na_2SiO_3

✚ The chemical formula for baking soda is NaHCO_3

✚ The chemical formula for POP is $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$

✚ The chemical formula for gypsum is $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

✚ The bond formed in the reaction Between water and hydrogen ion is called...?
=> Co-ordinate bonding (Dative bonding).

✚ The catalyst used in hardening vegetable oil is Nickel

✚ The catalyst used in the contact process is vanadium (V) oxide ... V_2O_5

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✚ The catalyst used in the Haber process is finely divided iron

✚ Gases with equal volume at s.t.p tends to have equal number of moles is law proposed by?
=> Avogadro

✚ The molecular shape of trichloromethane is ...?
A. Trigonal B. Tetrahedral C. Trigonal planar D. Trigonal pyramidal

=> B

✚ What's the IUPAC name for baking soda is sodium hydrogen carbonate

✚ Inorganic benzene is called Borazine

✚ The chloroform formerly used as anesthesia is trichloromethane CHCl_3

✚ The white colour in toothpaste is mainly from ...?

Titanium Dioxide (TiO_2).

✚ The base used in making toothpaste is $\text{Mg}(\text{OH})_2$

✚ $\text{Mg}(\text{OH})_2$ is also known as milk of magnesia, which is used as an antacid

✚ The acid found in bee sting is Formic acid (Methanoic acid)

✚ The acid found in vinegar is acetic acid (ethanoic acid)

✚ Methanol is also known as wood spirit

✚ Wasp sting are treated with acid like vinegar because wasp sting is alkaline unlike bees and ants .

✚ 95% of ethanol with water is known as rectified spirit

✚ A mixture of iodine in ethanol is known as tincture of iodine

✚ NO_2 is commonly known as brown gas

✚ The IUPAC for laughing gas is Dinitrogen (I) oxide . Also called nitrous oxide

✚ The chemical formula for laughing gas is N_2O

✚ ZnO is also known as philosopher's wool

✚ The following are the subshell and their shapes

S- spherical

P - dumbbell

d- double dumbbell

F - complex

✚ Azimuthal quantum number is also known as subsidiary or angular quantum number

✚ The energy level or shell of the electron is determined by?

=> Principal quantum number

✚ The shape of orbital is determined by?

=> Azimuthal quantum number

✚ The bond angle in water is 104.5°

✚ The shape of water molecule is bent or v shape

✚ The bond angle for ammonia NH_3 is 107°

✚ The bond angle in methane is 109.5°

✚ The geometry for methane is tetrahedral

✚ The hybridisation of methane is sp^3

✚ The hybridisation of ethene is sp^2

✚ The hybridisation of alkynes sp

✚ The hybridisation of alkenes and benzene is sp^2

✚ Hybridisation of graphite is sp^2 while Hybridisation of Diamond is sp^3

✚ Graphite conducts electricity due to the presence of free mobile electrons

✚ Diamond gives a bright shiny/sparking appearance due to its high refractive index

✚ Electrons fill lowest energy orbitals first is given by?

=> Aufbau principles

✚ Aufbau principles is also known as building principles

✚ No two electrons have identical quantum numbers is stated by who?

=> Pauli . The principle is called Pauli exclusion principles

✚ Orbitals of the same energy are filled singly first before pairing was stated by....?

=> Hund , which is called the Hund's rule.

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✚ Brown Ring Test is Used to detect nitrate ions (NO_3^-).

Test involves FeSO_4 solution and concentrated H_2SO_4 . 🔥

✚ Pentane (C_5H_{12}) is a liquid at room temperature.

✚ The Most abundant noble gas in air is Argon (Ar) ✓

✚ The Most abundant noble gas in the universe is Helium (He) ✓

✚ Argon (Ar) is Used in electric lamps to prevent oxidation of the filament .

✚ Krypton (Kr) + Xenon (Xe) are Used in photographic flash tubes for high-speed photography.

✚ Neon (Ne) is Used in aerodrome beacons and advertising signs.

✚ Neon has a Red-orange glow has high fog-penetrating power due to long wavelength. dangerous.

🔥 Helium + Oxygen (Heliox) mixture used by divers to prevent nitrogen narcosis. ✓

✚ Radon (Rn) is a Radioactive noble gas formerly used in cancer treatment (brachytherapy).

Now replaced by Cobalt-60 or Cesium-137 for safety.

✚ Synthetic Detergent (ABS) Alkyl Benzene Sulphonate.

✚ A graph of solubility (g solute per 100 g solvent) against temperature ($^\circ\text{C}$).

✚ Most solids become more soluble as temperature increases. ✓

✚ Solubility of Gases Increases with pressure — more gas forced into liquid (Henry's Law). ✓

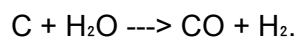
✚ Decreases with increase in temperature — gases escape faster when warm. ✓

👍 CO_2 stays dissolved in cold, high-pressure soda bottles.

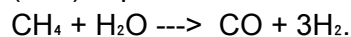
👍 When opened or warmed, gas (CO_2) escapes and drink goes flat.

✚ Water Gas [$\text{CO} + \text{H}_2$]

🔥 water gas is Produced by passing steam over red-hot coke.



✚ Synthetic Gas (Syngas) [$\text{CO} + 3\text{H}_2$] is produced by reacting steam with methane (CH_4) in presence of nickel catalyst (Ni).



✚ Graphite occurs naturally as plumago

✚ the shape of Graphite plate is hexagonal

✚ the shape of a diamond crystal is Octahedral

✚ the colour change of phenolphthalein in acidic medium is colourless

✚ graphite is a good conductor of electricity because of it's free mobile pi electron

✚ the impurities found in wood charcoal is sulfur

✚ the purest amorphous allotrope of carbon is sugar charcoal

✚ the salt used in glazing of pottery is NaCl sodium chloride

✚ the salt used in petroleum refining is zinc chloride ZnCl_2

✚ the salt used as an anti freeze is Calcium chloride CaCl_2

✚ the salt used in dyeing and calico printing is CaSO_4

✚ which of the following is a liquid at room temperature

A. Ethane

B. Propane

C Butane

D. Pentane ☒

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✚ Electronegativity increases across the period and decreases down the group

✚ Chlorine is the element with the most highest electron affinity

✚ Fluorine is the most electronegative element

✚ Silver mirror test is used to distinguish alkanals from alkanones

✚ Silver mirror test is positive with which functional group?

Ans: - CHO-

✚ Which of the following reagents will give a silver mirror on aldehydes

=> Tollens' Reagents

✚ The constituent of Tollens' Reagents is $\text{AgNO}_3 + \text{NaOH} + \text{NH}_3$

✚ Ionic size /atomic radius/electropositivity decrease s across the period and increases down the group

✚ screening effects increases down the group and remain constant across the period

✚ First ionization energy increases across the period and decreases down the group

✚ Brown ring test is used to detect the presence of nitrate ions NO_3^-

✚ H_2S (hydrogen sulfide) has a rotten egg smell

✚ Who's the father of chemistry?

=> Antoine Lavoisier

✚ Na_2CO_3 , sodium trioxocarbonate doesn't decomposed on heating

✚ Oxygen is produced industrially from the thermal decomposition of KNO_3 (potassium trioxonitrate)

✚ Which of the following hydrocarbon will show a more sooty flame?

=> Benzene/ C_6H_6

✚ The graph of solubility is concentration in mol dm^{-3} against _____?

=> Temperature

✚ What's the basicity of H_3PO_4 (phosphoric acid)

=> 3

✚ What's the basicity of H_3PO_3 (Phosphorus acid)

=> 2

✚ The bonding which results to no new substance is

=> Metallic bonds

✚ The salt used in glazing poultry is $NaCl$. $NaCl$ is also known as table salt or rock salt.

✚ Xenon and krypton is used in photographer's flash tube for taking high speed pictures

✚ Argon is used in gas filled electric lamps because it helps prevent the oxidation of the lamp filament

✚ The most abundant noble gas in the atmosphere is argon

✚ The most abundant noble gas in the universe is helium

✚ The most abundant elements in the atmosphere is nitrogen

✚ The most abundant elements in the earth crust is oxygen

✚ The most abundant metal in the earth crust is aluminium

✚ The most abundant metalloid in the earth crust is silicon

✚ The densest non metal is iodine

✚ The second most abundant elements in the earth crust is silicon

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✚ The process of water purification is

Coagulation > sedimentation > filtration > disinfection

✚ Copper is the element which is an anti fungal in nature , used as a fungicide in the form of $CuSO_4$

✚ Temporary hardness of water is caused by either

$Mg(HCO_3)_2$ or $Ca(HCO_3)_2$

Hydrogen trioxocarbonate of calcium or magnesium

✚ Temporary hardness of water can be removed by boiling or by the used of calculated amount of slake lime

✚ Stalagmites grows upward

✚ Stalactites grows downwards

✚ Permanent hardness of water is due to magnesium sulfate or calcium sulfate That's $CaSO_4$ or $MgSO_4$

✚ The impurities found in animal charcoal is calcium phosphate

✚ Liquid dispersed in liquid is emulsion

✚ Liquid dispersed in gas is Aerosol

✚ Solid dispersed in gas is called smoke

✚ Liquid dispersed in liquid is emulsion

✚ Liquid dispersed in gas is Aerosol

✚ Solid dispersed in gas is called smoke

✚ Solid dispersed in liquid is called Sols

🔥 Tests for proteins

✚ Biuret test : this requires no heating. Colour change is violet

✚ Millons test - colour change is brick red

PHYSICS (NOT FOR SALE).

✚ semi-conductor conduct better at higher temperatures explain why it has?

=> Negative temperature coefficient

✚ The addition of impurities to a pure semi-conductor in order to enhance conductivity is called Doping

- ✚ P-type semi conductors are semi conductors doped with Trivalent elements
- ✚ N- type semi conductors are semi conductors doped by pentavalent elements
- ✚ An A.C can be converted to D.C by the use of?

=> Diode

✚ When positive terminal is connected to P-type and negative terminal is connected to N-Type in a P-N junction diode , it's called FORWARD BIASING

- ✚ In forward biasing
- 🔥 Depletion region is small
- 🔥 Resistance is small
- 🔥 Current is high (Good conductor)

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- ✚ P-type semi conductor has deficit electrons
- ✚ N - type semi conductor has excess electrons
- ✚ An Ac generator can be converted to a DC generator by replacing Slip ring by split ring
- ✚ Split ring is also known as commutators
- ✚ Flemings left hand rule is also known as motor rule
- ✚ Flemings right hand rule is also known as Generator rule
- ✚ increasing light intensity only increases the number of photons not the energy or velocity.
- ✚ The complementary variables of Heisenberg uncertainty principles are
- => Energy and time
- => Momentum and position
- => Velocity and position

- ✚ The Compton's effect and photoelectric effect shows the particle nature of light
- ✚ Sound waves are longitudinal.
- ✚ Light waves are transverse.

- ✚ Sound requires a medium.
- ✚ Light travels fastest in vacuum.
- ✚ Wave speed equals frequency times wavelength.
- ✚ Pitch depends on frequency.
- ✚ Loudness depends on amplitude.
- ✚ Quality/ timbre depends on overtones or harmonics
- ✚ Reflection obeys equal angles law.
- ✚ Refraction occurs due to change in speed.
- ✚ Total internal reflection occurs from denser to rarer medium.
- ✚ Dispersion splits white light into colours.
- ✚ Interference occurs when waves overlap.
- ✚ Beats is as a result of interference
- ✚ Diffraction is the spreading of waves , this explains why we can hear through corners but we cannot see through corners

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- ✚ Plane mirror forms virtual, Erect and laterally image with the same size as object.
- ✚ Convex lens converges light, hence called converging lens .
- ✚ Concave lens diverges light, hence called diverging lens
- ✚ Myopia is corrected with concave lens.
- ✚ Hypermetropia is corrected with convex lens.
- ✚ Astigmatism is corrected by cylindrical lens
- ✚ Presbyopia is corrected by Bifocal lens
- ✚ Cloudiness of the eye lens at old age is due cataract
- ✚ Conventional current flows positive to negative.
- ✚ Fuse are always connected in series
- ✚ Magnetic field lines never intersect.
- ✚ Magnetic monopole doesn't exist

✚ Light behaves as particles and waves.

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✚ Generators convert mechanical to electrical energy.

✚ Motors convert electrical to mechanical energy.

✚ Turning effect increases with distance from pivot.

✚ Barometers measure atmospheric pressure.

✚ Manometers measure gas pressure.

✚ Buoyant force equals weight of displaced fluid.

✚ Latent heat involves temperature constant during phase change

✚ Virtual images cannot be projected on a screen.

✚ Motors convert electrical to mechanical energy.

✚ Optical fibre works on total internal reflection.

✚ Prism disperses light into spectrum.

✚ Red light deviates least in dispersion.

✚ Violet light deviates most in dispersion.

✚ Red light has the highest speed

✚ Violet light has the lowest speed

Use the Trend for light passing through a glass prism below 👉

ROYGBIV

🔥 From left to right - deviation increases

🔥 From right to left - speed increases

Where

R - Red , O- Orange , Y- Yellow, G- Green, B- blue , I- indigo, V- Violet

✚ All right spectrum have the same speed in vacuum

✚ Proof plane is used to test for the presence of charge and transfer charge

✚ Angle of Dip is 90° at the pole and 0° at the equator

✚ Angle of Dip is also known as angle of inclination not ~declination~

✚ The angle between the magnetic meridian and geographic meridian is called angle of declination

✚ The angle of between the magnetic field and the earth horizontal is inclination also called angle of dip

✚ Virtual image cannot be caught on a screen

✚ Pinhole camera produces Real inverted and diminished image

✚ Pinhole camera is used to take photos of still objects

✚ The type of mirror used in surveillance is convex mirror.

✚ The instrument which gives pure notes when sounded is Tuning Fork .

✚ Which of the following is electrically charged

A. Alpha B. Gamma C. Neutron D. X-ray
=> Alpha ✓

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✚ The device used in storing charge is called capacitor

✚ When an Ebonite rod is rubbed with fur
=> the Ebonite rod acquires a net negative charge

=> while the fur acquires a net positive

✚ When a glass rod is rubbed with silk ,
=> The glass rod Acquires a net positive charge

=> While the silk acquires a net negative charge

✚ Dull black surface is a good absorber and emitter of heat than polished surface

✚ Alpha particles have the highest ionizing power, while Gamma rays have the lowest.

✚ Gamma rays are the most penetrating (stopped by thick lead), while Alpha particles are the least (stopped by paper).

✚ Alpha is a Helium nucleus (2He^4), Beta is a fast-moving electron ($-1e^0$), and Gamma is an electromagnetic wave.

✚ Virtual image can be formed by objects on a concave mirror only when the object is placed between the principal focus and the pole

✚ Bicycle pumps operate by what principle?
=> Boyle's law.

✚ The direction of induced EMF or current is best explained by LENZ law

✚ The magnitude of induced EMF or current is best explained by Faraday's law

✚ In simple harmonic motion
=> Velocity leads displacement by 90° or $\pi/2$
=> Acceleration leads displacement by 180° or π

✚ Acceleration leads velocity by 90° or $\pi/2$
✚ RMS value of current is = 0.707 times the peak current

✚ RMS value of voltage is = 0.707 times the peak voltage

At Resonance in an A.C circuit

=> Power factor is 1

=> Phase angle is 0

=> Current is maximum

=> Capacitive reactance equals inductive reactance ($X_c = X_L$)

=> Resistance is minimum

=> Resistance = impedance

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✚ The reciprocal of resistance is conductance

✚ The reciprocal of resistivity is conductivity

✚ The reciprocal of impedance is admittance

✚ The reciprocal of Reactance is susceptance

✚ Mechanical advantage is also known as force ratio

Okay

✚ Velocity ratio only depends on the geometry of the machine

✚ The upper fixed point of an unmarked thermometer can be determined by the hypsometer

✚ The melting and refreezing of ice due to the variation of pressure is called regelation

✚ The type of thermometer which doesn't require contact before measurement of temperature is called pyrometer

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